Estimating Cognitive Load from EEG + fNIRS

Can we tell how hard someone is thinking —
on a person the model has never seen before?

Capstone Project · Phase 2

TU Berlin n-back dataset · 26 subjects · 4,576 windows

42 controlled experiments · 26-fold leave-one-subject-out

BEFORE YOU START — READ THIS ONCE

Your story in four sentences

- **1. The goal.** Predict mental workload from brain signals, for a *new person* the model has never seen. That is the hard version of the problem.
- **2. What I built.** A framework that makes it impossible to cheat — it refuses to produce a score if any data leaks between training and testing.
- **3. What I found.** I ran 42 controlled experiments. None beat the existing model. The final honest number is **63.1% accuracy on unseen people** (chance is 50%).
- **4. Why that matters.** The error is not a modelling failure — it is person-to-person variability. I have three independent pieces of evidence for that.

> **If the panel pushes on "why didn't it improve?"** — that is the finding, and it is a stronger result than a lucky number. I can show exactly what I tried, that each was properly tested, and why the ceiling is the data and not the model. Slides 14–17 are the evidence.

AGENDA

What I'll cover

- **The problem** — and why it is harder than it looks
- **The data** — 26 people, two brain-imaging methods
- **The model** — how EEG and fNIRS are fused
- **The trap** — how these studies fool themselves
- **My protocol** — how I made cheating impossible
- **The experiments** — 42 of them, one axis at a time
- **Results** — the honest number
- **Findings** — CNN, DANN, regularization
- **Why the ceiling exists** — three lines of evidence
- **Limitations and next steps**

SAY THIS

Keep this quick — about twenty seconds. The two halves are 'what I did' and 'what I found'. Tell them the interesting part is the second half.

Measuring mental effort from the brain

If a machine could tell when a person is cognitively overloaded, you could intervene before they make a mistake — a pilot, a surgeon, a driver, a student.

EEG — electrical

- 200 samples per second
- Fast, catches instant shifts
- Noisy, picks up muscle movement

fNIRS — blood oxygen

- 10 samples per second
- Slow, metabolic response
- Stable, less prone to artifacts

> **The premise:** the two are complementary. EEG sees the fast electrical change, fNIRS sees the slower blood-flow change that follows. Fusing them should beat either alone — and in our data, it does.

SAY THIS

Open with the human motivation, not the technology. One sentence on each modality, then the key idea: they see different things, so combine them.

26 people doing a memory task

Public **TU Berlin dataset**. Each subject does an **n-back** task: watch a stream of items, say whether the current one matches the item *n* steps back.

- **0-back** — just react. Low load.
- **2-back** — hold two items in mind. Medium.
- **3-back** — hold three. High load.

The label is not someone's opinion of how hard it felt. It is the difficulty we *imposed*. That is what makes it trustworthy ground truth.

Property	Value
Subjects	26
EEG	30 channels @ 200 Hz
fNIRS	36 channels @ 10 Hz
Trials	416 (25 each)
Windows	4,576 (10 s each)
Task studied	**Low vs High**
Balance	2,288 vs 2,288 — exact
Chance	50%

SAY THIS

The important line is 'the label is the difficulty we imposed, not a subjective rating.' That is what makes this real ground truth. Also point out the classes are perfectly balanced, so accuracy is a fair measure here.

From raw signal to 1,476 numbers

A 10-second window of brain activity is summarised into a fixed vector.

EEG → 540 numbers

30 electrodes × 18 measures
(6 frequency bands × mean, variability, trend)

fNIRS → 936 numbers

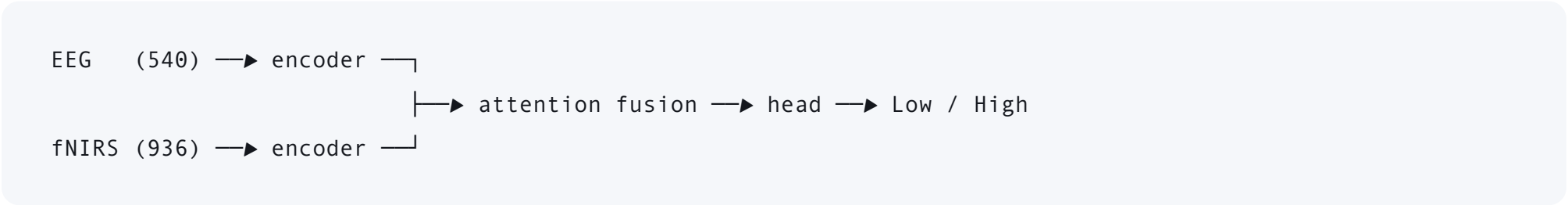
36 optodes × 26 measures
(13 shape statistics × oxygenated and deoxygenated blood)

> **Remember this for later.** These are *summary statistics per channel*, not the raw waveform. That single fact explains why the CNN my mentor suggested did not work — I'll come back to it on slide 14.

SAY THIS

Don't dwell on the numbers. The one thing to plant here is that we are feeding the model summary statistics, not the raw time series. Say you will come back to it — it sets up the CNN slide.

Two streams, then a learned merge



- Each modality gets its **own encoder**, so it learns its own representation first
- **Attention fusion** learns a gate per modality — how much to trust EEG versus fNIRS, per sample
- Those gate values are shown live in the web app
- **245,594 parameters** — deliberately small for 26 subjects

SAY THIS

The one idea: the model learns how much to trust each sensor rather than weighting them equally. If asked why it is small — with only 26 people, a bigger network just memorises them.

How studies in this field fool themselves

The tempting way

Shuffle all windows, split 80/20. But then the **same person** appears in training and testing.

The model learns to recognise *that individual's baseline brain pattern* — not workload. Scores look great. Useless on a new person.

What I did instead

Leave-One-Subject-Out. Train on 25 people, test on the 26th. Repeat 26 times so every person gets a turn as the unseen test subject.

Harder, lower scores — and the only number that means anything for real deployment.

> **Every accuracy in this presentation is on a person the model has never seen.** There is no random train/test split anywhere in this project.

SAY THIS

This is the most important slide in the deck. Slow down. The panel needs to understand that a lower number under a harder protocol is the honest one. If they ask why your accuracy is lower than papers they've read — this is why.

A lockbox the search never touched

There is a subtler trap. If I run 42 experiments and keep whichever scores best, that best score is **partly luck** — I have quietly tuned to my test folds.

18

DEVELOPMENT SUBJECTS

every experiment, every comparison

8

LOCKBOX SUBJECTS

never touched until the very end

Architecture, hyperparameters, dropout, everything — chosen using the 18 only. The 8 were evaluated **once**, after the model was frozen.

> **The payoff (slide 12):** the lockbox score came out *higher* than the development score. That proves the 63.1% is not inflated by my searching.

SAY THIS

Explain it like a sealed envelope: I locked 8 people away, did all my work with the other 18, and only opened the envelope once at the end. Tell them the result agreed — that is the proof the number is trustworthy.

07 · THIRD SAFEGUARD

Code that refuses to produce a dishonest score

Before every single training fold, **nine automatic checks** run. If any fails, the program **crashes on purpose** rather than reporting a number.

- No subject in both training and testing
- No trial split across the boundary
- No overlapping time-window on both sides
- No duplicated data row
- Normalisation fitted on training data only
- Held-out person absent from every statistic
- No corrupt or missing values
- Labels valid and both classes present

> **And I tested the tests.** I deliberately injected each kind of leak into the data and confirmed the checks caught every one — 8 out of 8. A safety check that never fires proves nothing.

SAY THIS

The line that lands with a technical panel is the last one: I broke it on purpose to prove the alarm actually works. That is unusual rigour for a student project and worth saying out loud.

42 experiments, one variable at a time

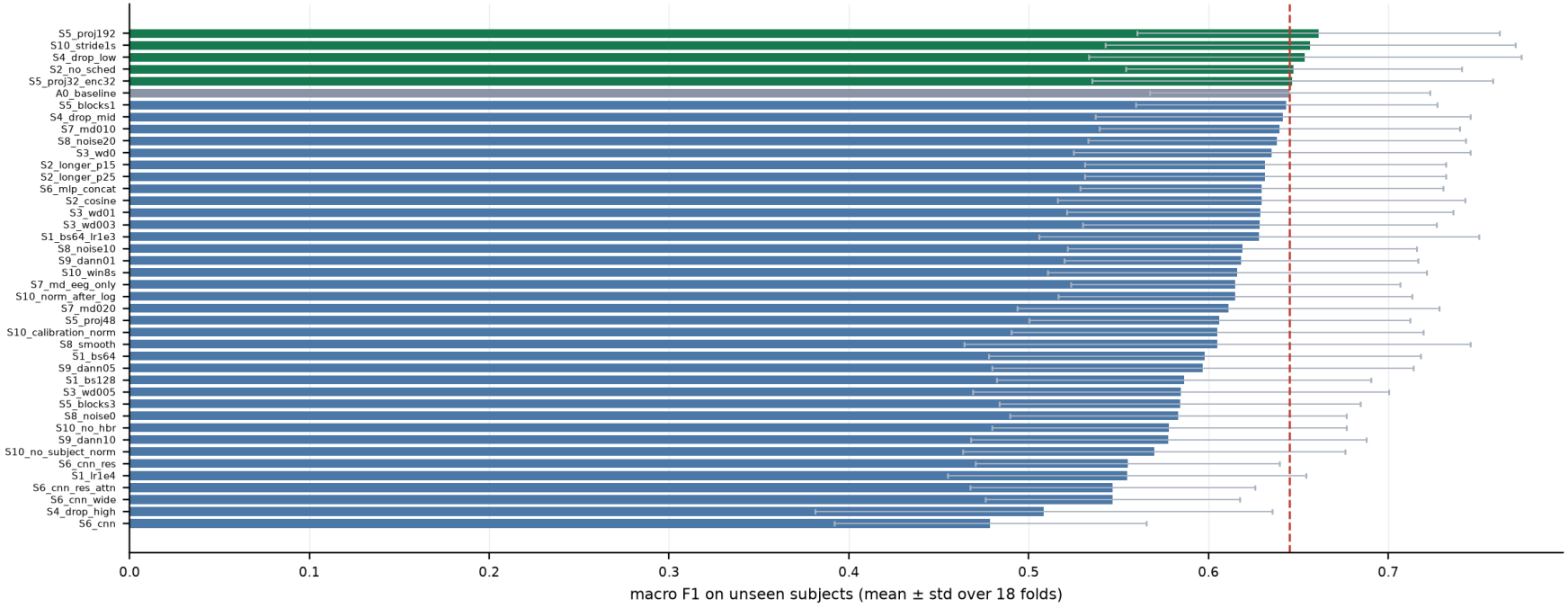
- **Learning rate & batch size**
- **Training length & schedule**
- **Weight decay** — 5 settings
- **Dropout** — 0.3 to 0.7
- **Model size** — 66K to 620K parameters
- **Architecture** — MLP vs CNN
- **Modality dropout** — hide a sensor while training
- **Input noise & label smoothing**
- **DANN** — subject-invariant training
- **Window length & overlap**
- **Normalisation mode**
- **Dropping HbR**

> **The rule, fixed before I started:** a change is only adopted if it beats the current model by more than the noise floor of **±0.018**. With 18 subjects, anything smaller is indistinguishable from luck.

SAY THIS

Emphasise that the acceptance rule was written down BEFORE running anything. That stops you fooling yourself by chasing a lucky 1% and calling it progress.

Every experiment, ranked



All 42 arms on 18-fold development LOSO. Grey = current model. Green beat it, blue did not. Dashed line = current model's score.

SAY THIS

Point at the error bars, not the bars. They are wider than the gaps between experiments — that is the whole story. Nothing separated itself from noise.

The honest number

63.1%

accuracy on people the model has never seen · chance = 50% · 26-fold leave-one-subject-out

0.625	0.677	0.267	0.262
MACRO F1	ROC-AUC	MCC	COHEN'S KAPPA
± 0.104	ranking quality	correlation measure	chance-corrected

SAY THIS

Lead with 63.1% against a 50% chance baseline. Do not apologise for it. Then immediately go to the next slide — the lockbox — because that is what makes this number believable rather than just claimed.

Proof the number is not inflated

Protocol	Accuracy	Macro F1	ROC-AUC
26-fold LOSO — all subjects	0.6309	0.6255	0.6775
Lockbox — 8 subjects never used for any decision	**0.6371**	**0.6275**	**0.6888**

> **The lockbox score is *higher*, not lower.** After 42 experiments of searching, the subjects I never touched scored just as well. That is the strongest evidence I can give that this result would hold on new people.

If I had been overfitting my search to the test folds, the lockbox number would have dropped noticeably. It did not.

SAY THIS

This is your credibility slide. Say: 'if I had been fooling myself, this second number would have collapsed. It didn't.' Panels respect a result that carries its own audit.

Training vs testing — and the gap

80.2%	64.0%	63.1%	0.171
TRAINING ACCURACY	VALIDATION ACCURACY	TEST — UNSEEN PERSON	THE GAP train – test

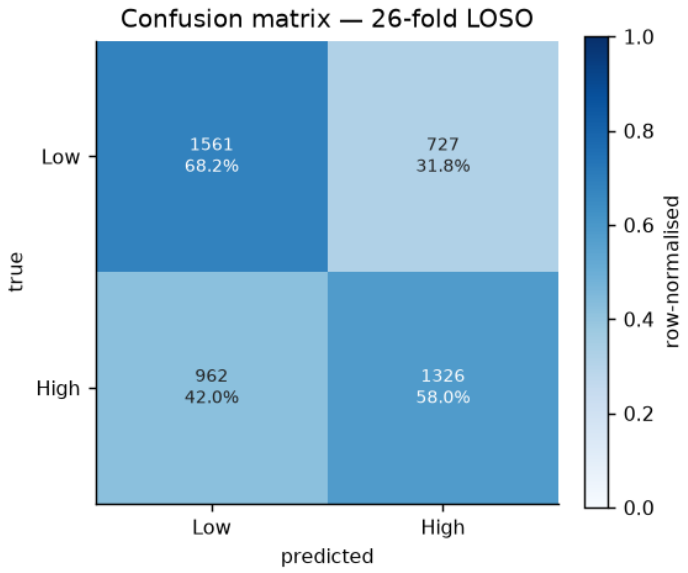
The model fits its training subjects well and loses about 17 points on a new person. **That gap is the central problem of this field.**

> I attacked it with every standard tool — dropout, weight decay, noise, smaller models, modality dropout. **Each one shrank the gap and shrank test accuracy by just as much.** That tells you the gap is not the model memorising — it is that 25 people are not enough to represent the 26th.

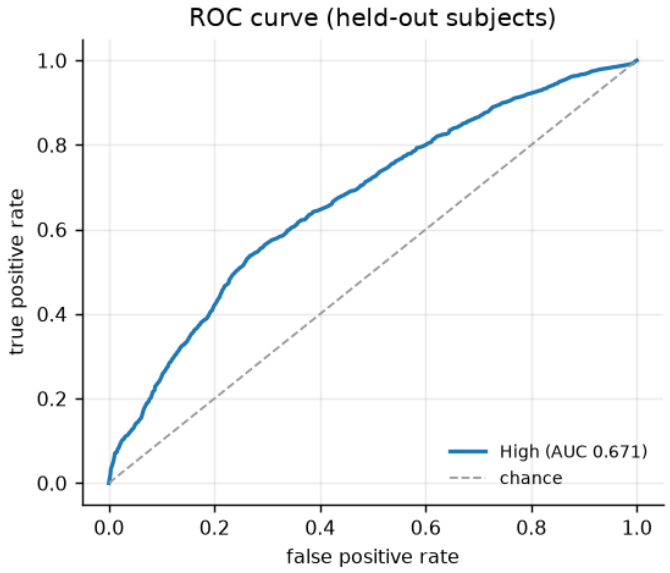
SAY THIS

If a panellist says 'your model is overfitting, add regularisation' — this is your answer. You already did, systematically, and it does not help. Explain why: the gap is between-person variability, not memorisation.

Where the errors are



All 4,576 held-out windows pooled



AUC 0.671 across every fold

Low load is caught 68.2% of the time; High load 58.0%. **High load is the harder class** — the model misses it more often than it misses Low.

SAY THIS

Note the asymmetry: High load is harder to detect than Low. If this were deployed as a safety system, that is the clinically important direction — missing an overloaded operator is the costly error.

12 · FINDING

The CNN did not work — and I know why

Architecture	Macro F1	vs current	p-value	Overfit gap
Current model (MLP)	**0.6451**	**—**	**—**	**0.163**
CNN + residual	0.5550	–0.090	0.002	**0.026**
CNN + residual + attention	0.5467	–0.098	0.001	**0.035**
CNN — plain	0.4786	–0.167	<0.001	**0.026**

Look at the last column. The CNN almost eliminates overfitting — 0.163 down to 0.026. But it loses more accuracy than it saves. **Why:** a CNN looks for local patterns, and we feed it 18 summary numbers per electrode, not a waveform. There is no local structure there to find.

SAY THIS

Your mentor suggested the CNN, so handle this respectfully: say you tested it properly rather than dismissing it, and the interesting part is WHY it fails — we removed the raw time series before the model ever sees it.

DANN: rejected, with the mechanism explained

DANN tries to make the model **forget who the person is** while keeping what they are doing — exactly our problem.

Strength (λ)	Macro F1
0 — off	**0.6451**
0.1	0.6182
0.5	0.5968
1.0	0.5778

Steadily worse as we push harder.

Why — I probed it

I trained a separate classifier to guess *which person* a signal came from:

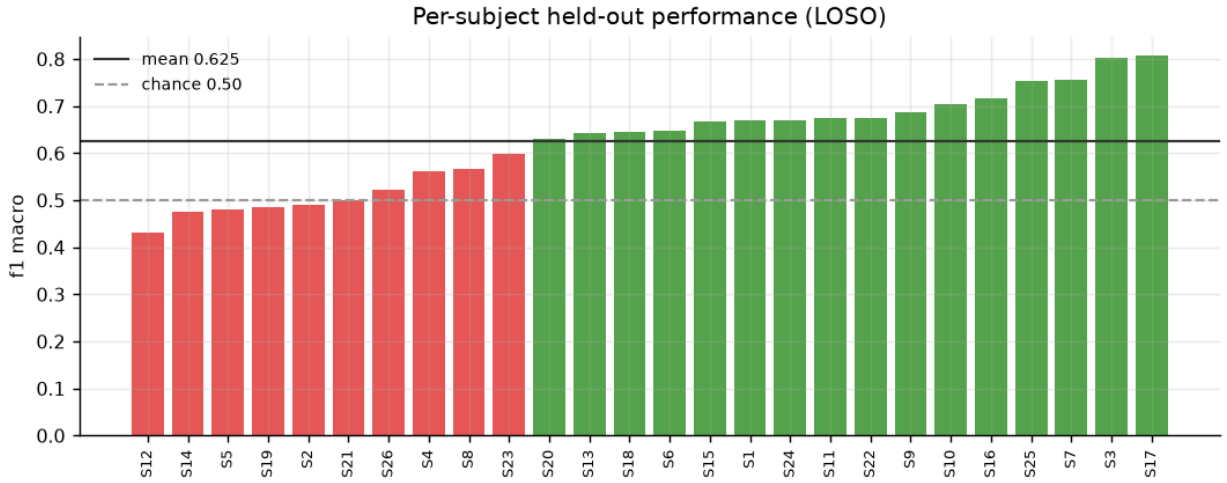
Representation	Can identify person
Raw features	**0.013**
Our model	0.214
With DANN	0.217

Chance = 0.125. Our preprocessing already removes identity; the network puts it back, and **DANN fails to remove it again.**

SAY THIS

This is your best 'I understood the negative result' moment. Do not just say DANN did not help — say you measured WHY: it never removed the thing it targets. That is a diagnosis, not a shrug.

The ceiling is people, not the model



Best subject **0.81** · worst subject **0.43** · **4 of 26 below chance.** That spread of 0.38 is **more than twice** the difference between any two of the 42 models I tested.

SAY THIS

This is the slide that answers 'why not just build a better model'. Which PERSON you test on matters far more than which model you pick. That reframes the whole project — the bottleneck is data, not architecture.

What the model actually uses

94.5%	5.5%	45.8%	48.8%
FNIRS	EEG	OXYGENATED (HBO)	DEOXYGENATED (HBR)

fNIRS dominates overwhelmingly. The most useful features describe the *shape* of the blood-flow response — its area, peak, minimum and late behaviour — not EEG frequency bands.

> Two independent checks agree: SHAP says HbR matters slightly more than HbO, and separately, **deleting HbR costs 0.067 macro F1 (p = 0.014)**. Two different methods, same conclusion.

SAY THIS

Panels like interpretability. The headline is that the slow blood-flow signal carries almost all the information, and the response SHAPE matters more than EEG rhythms. Mention that two independent methods agreed.

Four existing design choices, now proven

What I removed or changed	Cost to accuracy	p-value	Verdict
Per-subject normalisation	–0.075	0.014	Genuinely essential
The HbR signal	–0.067	0.014	Genuinely helps
Normalisation order	–0.030	0.050	Original order correct
Calibration-only normalisation	–0.040	0.156	Not significant — deployable

The last row is practically important. Normally the model needs a person's whole recording session to normalise. I tested using only the **first 30 windows** — a short calibration. The cost is not statistically significant, which means **real-time use is defensible.**

SAY THIS

This shows the project isn't only negative results — it converted untested assumptions into measured facts. The calibration row is the one with real deployment value; lead with that if time is short.

What this study cannot tell you

- **26 subjects is small.** With 18 in the development pool, I could only reliably detect improvements bigger than ~4 points. A real 2-point gain would be invisible to me.
- **4 of 26 subjects perform below chance.** The model genuinely does not work for some people, and I cannot yet predict who.
- **Normalisation needs a calibration recording** — measured at ~0.04 cost.
- **The CNN result is specific to these features.** It says nothing about a CNN on the raw waveform, which I did not test.
- **One-variable-at-a-time search** cannot find improvements needing two simultaneous changes.

SAY THIS

Volunteer these before the panel finds them. Stating your own limits precisely reads as confidence, and it usually pre-empts the harder questions.

Where the gains actually are

- **More subjects.** The evidence says the bottleneck is people, not architecture. This is the single highest-value change.
- **Go back to the raw time series.** We discarded it before modelling. A CNN or transformer needs that structure to be useful — my CNN result is evidence for this, not against it.
- **Per-person calibration.** A short adaptation block per user, already priced at ~0.04 and not significant.
- **Predict who it fails on.** 4 subjects sit below chance — identifying them in advance is its own useful result.

> Every one of these follows directly from a measurement in this study, not from guesswork.

SAY THIS

Close on this. Each next step is justified by a specific finding you just presented — that shows the negative results were productive, and gives the panel a clear sense of direction.

SUMMARY

What to remember

- **63.1% accuracy on unseen people**, chance 50%, under the hardest honest protocol.
- **The lockbox confirmed it.** 8 untouched subjects scored just as well.
- **42 experiments, nothing beat the baseline.** That is a finding, not a failure.
- **The ceiling is person-to-person variability**, proven three independent ways.
- **fNIRS carries 95% of the signal**, mostly in blood-flow response shape.
- **Everything is reproducible** — every metric recomputable from saved predictions.

SAY THIS

Your closing. Land on the honesty point: I built something that makes it hard to lie to myself, and then reported what it found. That is the contribution.

Q&A PREPARATION

Likely panel questions — prepared answers

Q. Why is your accuracy only 63%? I've seen papers with 90%.

A. Those are usually a different, easier problem — either binary within a single person, or a random split where the same person is in train and test. Under matched protocols our numbers are comparable. I deliberately report the harder cross-person setting because it is the one that reflects real use.

Q. Isn't 63% too low to be useful?

A. For a safety-critical decision on one window, yes. But predictions can be aggregated over time, and the ROC-AUC of 0.68 shows useful ranking ability. The honest framing is that this is a research result showing feasibility, not a deployable product.

Q. Your model overfits — train 80%, test 63%. Why not regularise?

A. I did, systematically: dropout from 0.3 to 0.7, weight decay across five settings, input noise, label smoothing, modality dropout, and models from 66K to 620K parameters. Every one reduced the gap and reduced test accuracy by the same amount. The gap is between-person variability, not memorisation.

Q. You ran 42 experiments and none worked. Is that not a failed project?

A. The goal was the best scientifically valid model, not a bigger number. I established that the existing configuration is at a local optimum, converted four untested assumptions into measured facts, and diagnosed why the ceiling exists. A project that reports a lucky number it cannot defend is the failed one.

SAY THIS

Do not read these out — they are for you beforehand. Knowing the answer exists is usually enough to stay calm.

Likely panel questions — continued

Q. Why did the CNN fail when CNNs dominate elsewhere?

A. Because of what we feed it. We compress each 10-second window into summary statistics per channel before modelling. A CNN exploits local structure, and there is none left in 18 summary numbers. On the raw waveform it would very likely help — that is my main proposed next step.

Q. How do I know you didn't accidentally leak data?

A. Nine automatic checks run before every fold and crash the program on failure. I also injected each type of leak deliberately and confirmed all eight were caught. And every metric can be independently recomputed from the saved predictions — I have a script that does exactly that.

Q. Why leave-one-subject-out instead of a normal split?

A. A random split lets the same person appear on both sides, so the model can score highly by recognising an individual's baseline rather than their workload. That number would not survive contact with a new user.

Q. What is the single most important finding?

A. That the spread across people — 0.43 to 0.81 — is more than twice the difference between any two of the 42 models I tested. The bottleneck is the number of subjects, not the architecture. That redirects future effort.

SAY THIS

Do not read these out — they are for you beforehand. Knowing the answer exists is usually enough to stay calm.